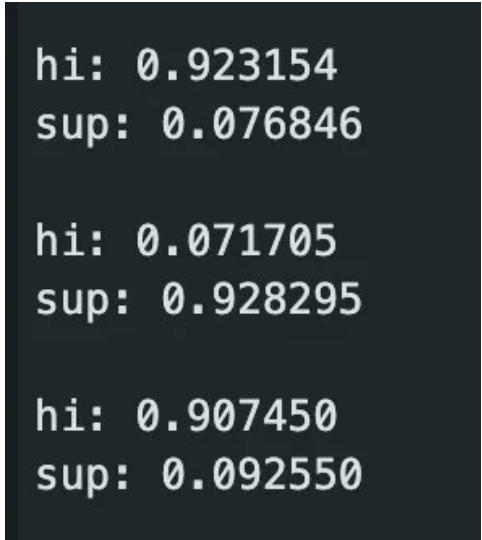


Software:

The bug was found in the Build and Train the Model portion of the notebook. While the classifier has been implemented with the proper softmax function (one-hot labels) as its output layer (hence used as a classifier), it was mistakenly compiled with MSE as the loss function and mean average error as the metric, thereby treating it as a regression problem. These were changed to categorical_crossentropy and accuracy, respectively. Since the change affected the metrics variable, I also needed to update the METRIC_TO_PLOT variable from "mae" to "accuracy" in order to make sure that the metric plot cell code works without throwing an error (the KeyError that occurred because there is no 'mae' key in the training history after the change). No modification was required to the loss plot cells, and I cleared out the previous cell outputs in order to make sure that they work properly again. After that, my network was successfully trained to 0.95 on the training set, 0.9 on the validation set, and 0.9 on the test set (as seen in the notebook); the TFLite conversion and model.h export also ran successfully. Finally, when I flashed model.h onto the board and tested gesture recognition in the Serial Monitor, I received "hi" and "sup" gestures with high confidence.

Hardware:



```
hi: 0.923154  
sup: 0.076846  
  
hi: 0.071705  
sup: 0.928295  
  
hi: 0.907450  
sup: 0.092550
```

hi: 0.071821
sup: 0.928179

hi: 0.174191
sup: 0.825809

hi: 0.223455
sup: 0.776545

hi: 0.051552
sup: 0.948448